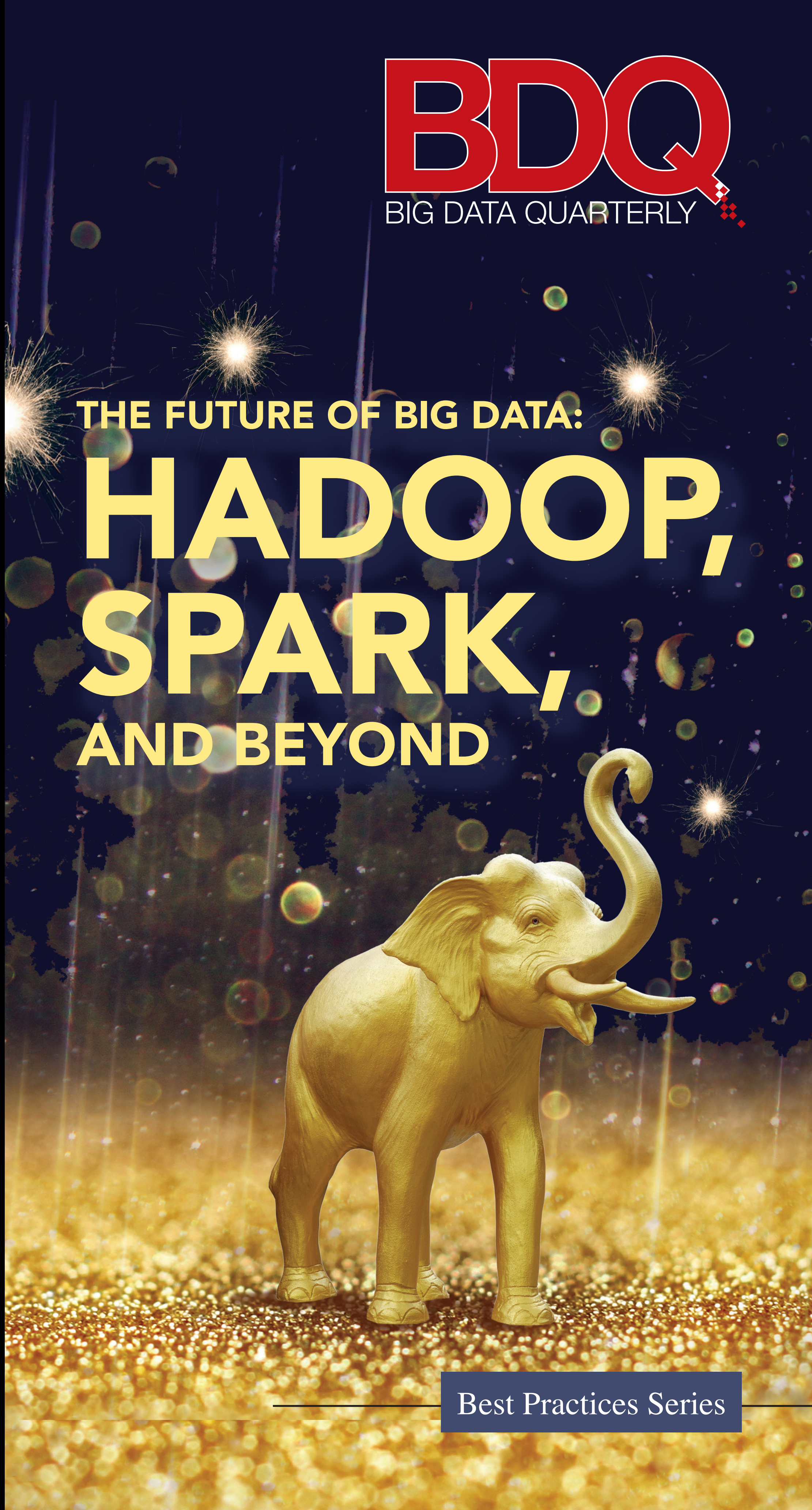
BDQ
BIG DATA QUARTERLY

THE FUTURE OF BIG DATA: HADOOP, SPARK, AND BEYOND



Best Practices Series

SEVEN WAYS TO UNLEASH THE POWER OF HADOOP

Best Practices Series

Hadoop burst on the scene several years years ago, offering a relatively cheap, open means for ingesting and analyzing large volumes of both unstructured and structured data. The big web-oriented companies (such as Yahoo and Google) that were dealing with massive quantities of data were the originators of Hadoop and the first to work with the technology, although in recent years, organizations of all types and sizes have embraced the framework. Hadoop is maturing, but along the way, challenges associated with delivering tangible gains for the business have emerged.

Data analytics and insights are where this value is being realized. Analytics—particularly across large or unstructured datasets—is a key competitive advantage of businesses going forward, and Hadoop helps enterprises overcome many of the costs and technical issues that have held these initiatives back in the past. "Hadoop is experiencing a level of adoption which indicates to us that its ecosystem will become prevalent in the area of analytics and BI applications for the corporate environment," according to a report by Robin Bloor and Rebecca Jozwiak of Bloor Research. "The Apache Hadoop stack is impressive, extensive and growing." Adoption has spread to organizations that have been able to "plug many of the gaps in Hadoop and enhance its capabilities, particularly in the areas of data analytics and BI."

The TDWI study also affirms that Hadoop implementations within enterprises are maturing, and increasingly, business value is being delivered. In the process, Hadoop projects are moving from smaller, departmental implementations to core IT oversight.

How can enterprises get the most out of Hadoop and translate these gains into business value? Here are some ways:

PROMOTE THE BUSINESS VALUE

The Hadoop open source framework has become a favorite of IT departments and data professionals, who see it as a way to quickly and cheaply analyze unstructured data such as log files and device data. Now, business leaders are beginning to recognize the potential business advantages the platform provides as well. Nine in 10 data executives and managers in a recent survey conducted by TDWI see Hadoop as a platform for increasing innovation within their organizations, along with support for analytics, data warehousing, and data scalability.

ADDRESS THE SKILLS GAP

Along with delivering tangible business gains, another challenge holding Hadoop back at this point is difficulties acquiring needed skills. As confirmed in the TDWI report, the leading barrier to Hadoop implementation is "inadequate skills or the difficulty of finding skilled staff (42%)." As many organizations are still in the early stages of adopting Hadoop as an enterprise framework, finding people who can build and understand its inner workings is still a challenge.

COLLABORATE WITH VENDORS AND OPEN SOURCE COMMUNITIES

Hadoop has been advancing—and becoming an easier-to-use, richer environment for enterprise data—because of the ongoing work of both associated vendors and open source community contributors. Enterprises interested in advancing Hadoop should work in close collaboration with these ongoing efforts.

MOVE TO THE CLOUD

There has been movement to the cloud, relieving many organizations of the headaches of building and maintaining clusters onsite. A number of vendors have released Hadoop as a service offering that appeals to enterprises wanting to expand their Hadoop environments without buying racks of servers and storage capacity. Managing large datasets can get expensive, and enterprises using cloud-based file management and analytics

Hadoop implementations within enterprises are maturing, and increasingly, business value is being delivered.

resources on a pay-as-you-go basis will reap cost advantages as well as flexibility. The cloud enables users to set up and access various configurations required for various workloads.

LOOK BEYOND MAPREDUCE

For the first few years of its existence, Hadoop has been closely paired with MapReduce. Emerging platforms such as Apache Spark offer ways to manage Hadoop Distributed File System (HDFS) data. "Despite its proven scalability, until the release of Yet Another Resource Negotiator (YARN), Hadoop was limited in its areas of application—it was used primarily for ETL and for archiving data to the HDFS," Bloor and Jozwiak observe. "Its two main constraints were that it ran just one task at a time and that software development on Hadoop was tied to the use of MapReduce. The October, 2013 release of Hadoop 2.0, which included YARN, eliminated these limitations entirely. As an immediate and expected consequence, Hadoop saw a significant increase in momentum." According to Bloor and Jozwiak, "the Hadoop story is an ugly-duckling-turned-swan story. Hadoop with MapReduce was not pleasing to the eye, but once MapReduce had been relegated to the position of optional component, suddenly it began to look much more appealing, and then enthusiasm for it in the large corporate computing environments accelerated."

LOOK BEYOND 'BIG DATA'

Hadoop is just as applicable to smaller sets of data as big data. Hadoop is adaptable for a range of data needs, not just big datasets. Hadoop serves data needs for a range of departments and functions across enterprises. Hadoop has long-term value, as well. For example, data warehouses will work in conjunction with Hadoop. There are also a range of additional data applications and areas that have long been part of the traditional relational database domain that Hadoop will support, including records management and content management systems.

ENHANCE—BUT NOT REPLACE—- EXISTING DATA ENVIRONMENTS

There are a number of reasons Hadoop has become part of the data operations of many enterprises. The sheer volume of data streaming through organizations has been overwhelming many traditional data environments. Relational database management systems that have been in place for years often cannot efficiently or effectively handle unstructured data. On the other hand, "Native Hadoop itself is not a data warehouse and cannot provide the optimized performance associated with such a product," say Bloor and Jozwiak. "Hence, it requires a complementary data warehouse capability or a similarly constructed solution engineered to work over HDFS." Similarly, 46% of the data professionals in the TDWI survey report they see Hadoop serving as a complementary extension of their data warehouses, the leading use case cited. Another 26% see Hadoop as the foundation for a data lake implementation. Typically, moving analytical jobs to Hadoop enables processing within a more inexpensive environment than traditional data warehouse or ETL settings.

Hadoop holds a great deal of potential, since it provides a highly agile and cost-effective way to bring big data closer to the business. But it is still a work in progress across most mainstream enterprises, and more employee training, executive education, high-level planning, and technology integration is required to realize this potential on an enterprise scale.

—Joe McKendrick

The Enterprise Data Hub in Financial Services

Three Customer Case Studies

The typical financial services adoption cycle for Apache Hadoop usually begins with one of the two most prominent operational efficiency and cost reduction use cases: data consolidation and multi-tenancy or full-fidelity analytics and regulatory compliance with a centralized data hub. However, an October 2013 study by Sand Hill Group found that only 11% of respondents had progressed beyond their first Hadoop project, and only 9% were using Hadoop for advanced analytics, despite the fact that 62% indicated that they anticipated advanced analytics becoming a top use case during the next 12 to 18 months.[1] With so many organizations seeking a reliable, real-time, and affordable big data solution, what is the barrier to full adoption and production?

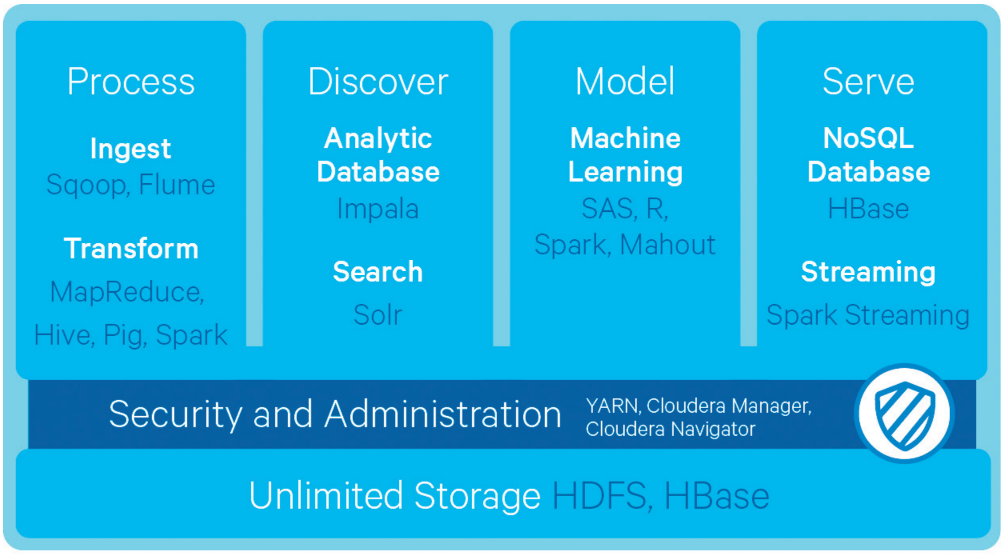
Unlike traditional data management and analytics platforms that are usually deployed as specialized systems with specific objectives, the central, open, and scalable nature of an enterprise data hub makes it more akin to a solutions engine for the financial services industry. By minimizing opportunity cost and emphasizing integration with a vast and growing ecosystem of relevant technologies and familiar applications, Cloudera is helping firms address their big data challenges today and maximize the option value of their data infrastructure for more advanced business objectives downstream. Bringing compute to all your data in service of an introductory use case actually enables, facilitates, and affords the opportunity for firms to quickly take advantage of new information-driven business competencies that were previously too expensive or complex for most enterprises: machine learning models for more effective and automatic fraud detection and prevention, recommendation engines to personalize the customer experience for up-sell and cross-sell opportunities, and a 360-degree view of the business for ad hoc exploration, experimental analysis, and advanced risk modeling.

[1] Graham, Bradley, and Rangaswami, M.R. *Do You Hadoop? A Survey of Big Data Practitioners.* Sand Hill group, October 2013.

A LEADING PAYMENT PROCESSING COMPANY AND FRAUD DETECTION

With the movement from in-person to online financial transaction processing, the number of daily transactions processed by a leading global credit card company has ballooned, causing increased susceptibility to fraud. By definition, fraud is an unexpected or rare event that causes significant financial or other damage—the effective response to which can be categorized, from the enterprise perspective, by detection, prevention, and reduction. In the financial services industry, anomalies usually occur because a fraudster has some prior information about how the current system works, including previous fraud cases and the fraud detection mechanisms, which makes building a reliable statistical model for detection very difficult.

In the case of this large credit card processor, despite an annual $1 billion budget for data warehousing, statisticians were limited to fairly simple queries on relatively small samples of data because anything more extensive would consume too many compute resources. In particular, data scientists within the



global information security group wanted faster query response and unconstrained access to better mine and analyze data in the relational database management system (RDBMS).

By deploying Hadoop as part of Cloudera Enterprise, this firm not only streamlined its data processing workflows and significantly reduced its anticipated costs by integrating all the jobs usually assigned to separate storage area network (SAN), extract-transform-load (ETL) grid, and data warehousing systems, but also immediately began examining data from a longer period of time and a greater variety of sources to identify more and different potentially anomalous events. To overcome latency, Apache Flume—Hadoop's service for efficiently collecting, aggregating, and moving large amounts of log data—can load billions of events into HDFS—Hadoop's distributed file system and primary storage layer—within a few seconds and analyze them using Cloudera Impala—Hadoop's massively-parallel-processing structured query language (SQL) engine—or even run models on streaming data using the in-memory capabilities of Apache Spark—the next-generation, open-source processing engine that combines batch, streaming, and interactive analytics on all the data in HDFS.

Today, the credit card processor ingests an average of four terabytes of data into its Hadoop cluster every day and is able to maintain thousands more across hundreds of low-footprint nodes for its fraud modeling. Shortly after deploying its enterprise data hub, the company was notified by a partner of a small incidence of fraud that had reportedly only been occurring for two weeks before detection. In response, the global information security group was able to run an ad hoc descriptive analytics model on its long-term detailed data in Hadoop—a task that would have been virtually impossible with traditional data infrastructure alone. By searching through the broader data set, the company found a pattern of the fraudulent activity over a period of months. This became the sector's largest detection of fraud ever, resulting in at least $30 million in savings.

Additionally, the company is using the data from its Hadoop cluster to create revenue-driving reports for merchants. Historically, certain monthly reports took two days to complete and required a large amount of processing power managed by a technical team. Now, the credit card processor is building a billion-dollar business by selling reports generated by combining much larger transaction data with purchase data from banks. The reports can be run in a matter of hours and overcome a latency issue merchants had faced when collecting data for customer segmentation and cross-sell analytics.

A TOP INVESTMENT BANK AND THE 360-DEGREE VIEW OF THE BUSINESS

With growing data volume and variety available for portfolio analysis, many investment banks struggle to figure out the best way to process, gain visibility into, and derive value from more data. Most rely on data sampling, which reduces the accuracy of their models and prohibits exploration.

The concept of a 360-degree view is usually associated with retail banks that want to use more data from more sources across multiple business units combined with on- and offline behavior trends to understand how to effectively and efficiently engage customers for greater loyalty and new selling opportunities. However, a broad, informed, real-time view of the business is not necessarily limited to customer happiness and marketing metrics. Combining related or even disparate data sets can reveal patterns, correlations, or causal relationships that, when translated into opportunity or risk, can provide investment banks with a valuable head start over other firms.

At a leading wholesale bank, competitive advantage is directly related to not only the quantity and quality of its data but, perhaps more importantly, the flexibility to investigate the relevance and relationship of insights to outcomes. The firm, which reported client assets under management in the trillions of dollars in 2013, balances not only its own market and investment data, but also relies on custom algorithms to draw actionable insights from public and policy information, macroeconomic data, client profiles and transaction records, and even web data—essentially always seeking to go one click down on any individual observation.

The investment bank's data scientists wanted to put very large data volumes to use for portfolio analysis, but the traditional databases and grid computing technologies they had in-house would not scale. In the past, IT would create a custom data structure, source the data, conform it to the table, and enable analysts to write SQL queries. This process was extremely precise and time-consuming. Often, when the application was handed off to the business, the analyst would indicate that the project did not deliver on the original request, and the application would go unused and be abandoned.

As a first big data proof-of-concept with Cloudera, the bank's IT department strung together 15 end-of-life servers and installed CDH, Cloudera's open-source distribution of Apache Hadoop, loaded with all the company's logs, including a variety of web and database logs set up for time-based correlations. With so much data online and available in Hadoop, the bank was able to explore its investment operations at petabyte scale from all angles for the first time. Because Hadoop stores everything in a schema-less structure, IT was able to flexibly carve up a record or an output from whatever combination of inputs the business wanted, and results could be delivered to the business on demand.

As a Cloudera Enterprise customer, the investment bank no longer relies on sampling, meaning its portfolio analysis is run at a much larger scale, delivering better results. Hadoop can search through huge volumes of data and run pattern-matching for every single imaginable attribute. A user does not have to know what he or she is looking for—just let the software and models detect patterns and then follow up with further investigation.

The time-based correlations over log data that are powered by an enterprise data hub allow the bank to see market events and how they correlate with web issues and database read-write problems with an unprecedented level of completeness and clarity. For instance, the company has access to an event's entire traceability in real time, in terms of who did what, when, and how, what caused the issue, and what kind of data was being transacted. The bank can tie front-office activities with what is going on in the back office and which data is causing unexpected results. In the past, figuring out what caused a system to perform incorrectly would take months and could cost the business plenty.

With Cloudera, the company can now figure out and solve problems as they happen, or even prevent them before they happen. Furthermore, advanced analytics tools deployed as part of the enterprise data hub also provide the bank's financial advisers with customized recommendations for clients to sell or buy stocks based on information gathered in real time on current positions and market conditions—essentially monetizing Hadoop's capabilities delivered and supported by Cloudera Enterprise: Data Hub Edition.

A LARGE INSURER AND FINANCIAL PRODUCT PERSONALIZATION

With the proliferation of sensors, mobile devices, nanotechnology, and social apps, individuals are more inclined than ever to monitor and passively or actively share data about their day-to-day behaviors. Insurers, who have historically competed on general pricing or via broad, expensive marketing campaigns, want to differentiate their coverage options by customizing plans based on information collected about the individual's lifestyle, health patterns, habits, and preferences. However, traditional databases cannot scale to the volume and velocity of real-time, multi-structured data required for policy personalization. An enterprise data hub enables real-time storage and stream processing for a competitive pay-for-use insurance model.

One of the largest personal insurance companies in the United States was initially founded as part of a national department store chain in the early-1930s. Over its more than 80 years in operation, the company has collected massive quantities of data, much of which was never digitized, and most of which was unstructured document content. As the insurer began to transition its historical and current policy data into online records and attempt to run programs that correlated such external data as traffic patterns, socioeconomic studies, and weather information, the IT department found that the systems would not scale to accommodate such variety of formats and diversity of sources.

A primary example of the challenge faced by business analysts was graph link analysis. For instance, they could look at data from a single U.S. state at a time—with each state's analysis requiring about a day to process—but could not run analytics on multiple states, no less all 50 states, at once. Although new data systems were being put in place to capture and prepare data for reporting and business intelligence, they were primarily aligned to marginally improve on old approaches to data management, which separated data types and workloads into distinct silos.

With a first objective of speeding up processing times and consolidating its disparate data sets to achieve more scalable analytics, this leading insurance company built an enterprise data hub with Cloudera Enterprise. Its centralized Hadoop implementation spans every system across the entire company to break down data silos and provide a single, comprehensive view of all its data. The three main technical cases for adopting Hadoop were flexible and active data storage, integrated and efficient ETL, and applied statistics and computation.

The insurer brought together customer account information, public economic and social studies, and telemetric sensor data in its initial Hadoop cluster. Some of these data sources had never been brought together before, and much of the historical data, which was newly digitized, could not be analyzed in tandem with external sources prior to landing in Hadoop. Today, the company's enterprise data hub is integrated with its incumbent mainframes and data warehouses—it was designed specifically to complement, not replace, existing infrastructure.

Now that it can run descriptive models across historical data from all 50 states using Apache Hive—open-source software that makes transformation and analysis of complex, multi-structured data scalable in Hadoop—the insurer is experiencing an average 7500% speed-up on analytics and seeing even better results with Impala. Unburdened by data silos, its analysts and data scientists are building predictive models that help the business customize products that are better aligned to the individual behaviors and risks of each customer, tune pricing of insurance plans more precisely to maximize lifetime value, and develop differenti¬ated marketing offers that communicate value for the most appropriate cross-sell and up-sell opportunities without diminishing margins.

BIG DATA AND AN ENTERPRISE DATA HUB

When information is freed from silos, secured, and made available to the data analysts, engineers, and scientists who answer key questions about the market—as they need it, in its original form, and accessed via familiar tools—everyone in the C-suite can rest assured that they have a complete view of the business, perhaps for the first time. For financial services firms, overcoming the frictions related to multi-tenancy on compliant and secure systems is the gateway to advanced big data processes: machine learning, recommendation engines, security information and event management, graph analytics, and other capabilities that monetize data without the costs typically associated with specialized tools.

Today, the introduction of an enterprise data hub built on Apache Hadoop at the core of your information architecture promotes the centralization of all data, in all formats, available to all business users, with full fidelity and security at up to 99% lower capital expenditure per terabyte compared to traditional data management technologies.

The enterprise data hub serves as a flexible repository to land all of an organization's unknown-value data, whether for compliance purposes, for advancement of core business processes like customer segmentation and investment modeling, or for more sophisticated applications such as real-time anomaly detection. It speeds up business intelligence reporting and analytics to deliver markedly better throughput on key service-level agreements. And it increases the availability and accessibility of data for the activities that support business growth and provide a full picture of a financial services firm's operations to enable process innovation—all completely integrated with existing infrastructure and applications to extend the value of, rather than replace, past investments.

However, the greatest promise of the information-driven enterprise resides in the business-relevant questions financial services firms have historically been unable or afraid to ask, whether because of a lack of coherency in their data or the prohibitively high cost of specialized tools. An enterprise data hub encourages more exploration and discovery with an eye towards helping decision-makers bring the future of their industries to the present:

How do we use several decades worth of customer data to detect fraud without having to build out dedicated systems or limit our view to a small sample size?

What does a 360-degree view of the customer across various distinct lines of business tell us about downstream opportunity and risk?

Can we store massive data on each customer and prospect to comply with regulatory requirements, secure it to assure customer privacy, and make it available to various business users, all from a single, central point?

ABOUT CLOUDERA

Cloudera is revolutionizing enterprise data management by offering the first unified Platform for big data, an enterprise data hub built on Apache Hadoop. Cloudera offers enterprises one place to store, access, process, secure, and analyze all their data, empowering them to extend the value of existing investments while enabling fundamental new ways to derive value from their data. Cloudera's open source big data platform is the most widely adopted in the world, and Cloudera is the most prolific contributor to the open source Hadoop ecosystem. As the leading educator of Hadoop professionals, Cloudera has trained over 40,000 individuals worldwide. Over 1,600 partners and a seasoned professional services team help deliver greater time to value. Finally, only Cloudera provides proactive and predictive support to run an enterprise data hub with confidence. Leading organizations in every industry plus top public sector organizations globally run Cloudera in production.

CLOUDERA
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MarkLogic

The Rise of Data Warehouses

PRIOR TO 1990, enterprises generally used the same relational database management system (RDBMS) platform for all their data, no matter what type of operations or analysis needed to be performed. But as the pace and scale of data grew, new architecture models began to be deployed to use technology in different ways based on different data and different business needs: the Enterprise Data Warehouse (EDW), for example, was introduced to create a cross-line-of-business analytical view, separate from online transactional systems.

With the rise of the EDW, organizations finally had an integrated view of their data to support management decisions. However, implementation and maintenance of the EDW remains heavily dependent on resource-intensive extract, transform, and load (ETL) processes for moving data around. And, because most EDWs are still based on RDBMS technology, they struggle to incorporate unstructured information, keep their database schemas current with the onslaught of new data sources, and scale without breaking the bank.

ENTER HADOOP

With EDW processes constrained by RDBMS limitations, Hadoop is a natural fit for helping with data management in the big data era—it delivers economies of scale for high-volume data storage and batch processing, along with flexibility to take in any shape of raw data including unstructured information.

Because of these advantages, many organizations have used Hadoop to extend their EDW architectures, for example, by:

- Spilling over data from the EDW into Hadoop for batch analytical processing, separate from existing RDBMS-based analytic tools
- Using a Hadoop-based "data lake" for ingest of information, which can then be used for analytical processing or to feed the EDW

However, despite Hadoop's apparent value in a traditional EDW architecture, organizations are still left with a number of challenges because:

- Hadoop is not a database, so RDBMSs are still required to support real-time applications
- Hadoop requires hard-to-find expertise to configure, deploy, and manage
- Security and governance limitations in Hadoop prohibit deployment in mission-critical environments
- There is still a lot of ETL time and cost, as RDBMSs remain in the picture

DERIVING BUSINESS VALUE FROM DATA

Industry analysts find that organizations face challenges in getting meaningful insights from the increasing amount of data they are ingesting and storing in a Hadoop ecosystem.[1] Not only that, but these enterprises are still stuck with a separation of powers—Hadoop (within or even in place of an EDW) for analytics running separately, downstream from operational activities.

Getting true value from analytics means being able to operationalize what you analyze, in real time. It means supporting analytics on the same data that you use to run your business—without waiting on ETL processes and without sacrificing performance. And, it's possible today.

INTRODUCING ENTERPRISE NOSQL

The MarkLogic Enterprise NoSQL platform complements Hadoop by addressing its limitations, and brings the big data promise to your operational workloads by supporting real-time discovery alongside mission-critical, run-the-business applications. It combines all the flexibility and cost-effectiveness of a scale-out NoSQL database along with built-in search and semantics, application services, and the enterprise-grade features your organization needs to ensure data reliability and security.

In addition to supporting numerous other storage types, MarkLogic can leverage Hadoop as a storage layer, allowing for the creation of a multi-tiered storage infrastructure. And, MarkLogic integrates with Hadoop as a compute layer for batch loading and processing. Organizations benefit from the power of MarkLogic and the cost-effectiveness of Hadoop to handle mixed operational and analytic workloads, perform simultaneous read and write during data discovery, use real-time alerting to react immediately to important events, create workflow based on analytical results, and cost-effectively manage data throughout its lifecycle.

The schema-agnostic MarkLogic platform lets you ingest and use the data format that makes the most sense for your data (JSON, XML, RDF, documents, binaries, geospatial, etc.). It indexes all data and content upon ingestion to support immediate information discovery and exploration—and ongoing search, query, and alerting. Data quality, context, and lineage are managed within the platform by leveraging built-in semantics and bitemporality, and keeping information about the data (metadata) close to the data itself. And, because MarkLogic is a complete database management system with ACID transactions, HA/DR, and government-grade security, it supports the real-time operational applications you need to run your business.

By combining MarkLogic and Hadoop, organizations no longer need to deploy separate EDWs and operational data stores. They get fewer moving parts—that do more.

MARKLOGIC www.marklogic.com

1 See, for example, Gartner Inc., *Architecture Options for Big Data Analytics on Hadoop*, 1 July 2015.

MAPR

Real-Time Hadoop Keeps Getting More Real

Many interesting topics related to Hadoop are bubbling to the top of the big data world. Future Hadoop requirements will likely include capabilities like real-time, interactive analysis, and security and data governance. Apache Spark, SQL-on-Hadoop engines, and NoSQL databases all promote real-time processing and interactive analysis in Hadoop. And, numerous third-party vendors have made strong pushes to address security and data governance in Hadoop. Let's take a look at each of these areas.

Real-time processing in Hadoop can be viewed from three separate perspectives. The first is about the data. Real-time data access means the data is available for processing as soon as it enters the system. The second is about applications, which is addressed by integrating NoSQL databases with Hadoop. The third is about analytics, where the ability to run queries with minimal data preparation is important when your data keeps moving. Real-time analytics provides the interactivity you need for analytical refinement and also uses the latest data to give you the most up-to-date outputs.

Real-time data is often handled via real-time stream processing. Typically, Apache Kafka combined with Apache Storm or Spark Streaming comprise the stack. Kafka handles the large-scale delivery of messages across systems, while Storm and Spark Streaming run operations on that streamed data. This stack can be used to calculate aggregates on the fly or to flag specific anomalous values that might indicate fraud or failure. Oftentimes, that incoming and processed data is stored in Hadoop for retrospective analysis, such as for creating predictive models. MapR Technologies takes real-time data further by providing a POSIX NFS interface and a fully read/write file system. With those features, MapR customers can load data as easily as copying files to a regular disk drive, which is then immediately accessible by Hadoop tools. These real-time data features in MapR also complement Spark, Kafka, and Storm well for further enabling real-time in Hadoop.

Real-time applications built on NoSQL databases will continue to be a powerful component of a Hadoop-based deployment. Integrations range from direct replication to Hadoop, to API connectivity, to direct I/O calls into the HDFS API. All of these efforts are intended to reduce the latency of copying live, operational data to a separate Hadoop cluster. MapR Technologies solved this problem by embedding a NoSQL database, MapR-DB, into the core platform. Any data written to MapR-DB, which uses the HBase API and the same wide-column data model, is immediately accessible to Hadoop tools. With its many automated optimizations, MapR-DB administrators don't have to shut down the system for disk and data cleanup tasks like compactions and "anti-entropy."

For real-time analytics, analysts should be able to query data sooner than later. Any technology that can minimize the IT intervention, like schema building, can help improve real time. Apache Drill, supported in the MapR Distribution, is one such technology because it reads non-relational data formats like JSON and CSV (plus other formats like Parquet) and instantly sets up a schema for use by SQL-based tools. This lets analysts and data scientists explore data in a self-service manner without the delays of setting up a schema that most relational database tools require.

For security and data governance, standards are still yet to be set. The MapR approach to security is to make it as powerful and flexible as possible at the platform level. For example, in addition to Kerberos integration support, MapR uses Linux Pluggable Authentication Modules (PAM) for the widest user registry support. MapR Access Control Expressions (ACEs) on MapR-DB give granular and easy-to-manage access controls. Drill views provide field-level access controls on unstructured files. Built-in auditing lets you log all data access, not only for understanding user behavior, but also for regulatory compliance. Encryption is also available, both for data-in-motion and at rest (via many MapR Advantage Partners).

Data governance in Hadoop is important with so much critical data in Hadoop. More sophisticated approaches are available from vendors with a history of delivering data governance solutions beyond Hadoop. But in many cases, specific policies can be easily addressed by platform-level features in MapR that are not found in other distributions. These features help with easier data integration, data lineage, retention and purging, and auditing. MapR volumes and snapshots, for example, can be used for supporting lineage and retention/purging policies.

If real-time processing, interactive analysis, and security and data governance are important to you, then be sure to look at MapR Technologies for your Hadoop, Spark, and NoSQL needs.

MAPR
www.mapr.com

The Key to Turning Hadoop Into an Enterprise-Grade Analytics Platform

Despite increased IT spending, the majority of Hadoop big data projects are still in the development lab. Enterprise data is going into the Hadoop data lake but organizations are often not generating value from it. That's a challenge that organizations need to overcome. Companies that are able to create more widespread access to the power of Hadoop across their organizations stand to reap huge gains.

HADOOP IN ACTION—CASE STUDY

For example, at Actian we worked with a leading bank that found by leveraging Hadoop they were able to process 20 times more data than was possible with their previous database. This bank could now query 200 billion rows of data and invest $100 billion float in just 28 seconds—as opposed to the 3 hours it once took—while instituting multi-level risk exposure analysis for controls and regulatory compliance across the organization. But in order to enjoy those benefits they also had to make specific investments.

Hadoop needs to be made accessible by the simple but powerful SQL. Despite the expensive skillsets, long and error-prone implementation cycles, lack of native support for popular BI tools and inadequate execution times, Hadoop remains a powerful framework for big data and a vital technology for the data-driven business.

Most businesses already have a conduit for data and analytics using BI tools: a business analyst. But in that conduit also lies the barrier, as most analysts rely on SQL, the lingua franca for managing relational database systems, to query data and perform analysis. Until now there has been little success in using SQL to access data in Hadoop, as most SQL in Hadoop products on the market have very limited SQL capabilities. The appetite for Hadoop-driven BI is there, but the gap between the data itself and those actionable results must be bridged.

HADOOP'S OPPORTUNITY

Traditional architectures inherently don't provide the price or performance required by new workloads. Given its ability to handle large data sets at a fraction of the cost, Hadoop offers a huge opportunity for enterprises to capture and derive insights from big data and ultimately drive transformational business outcomes. However, SQL has long been the standard trade language for those who work with data and databases. The benefits are simply too great to overlook. Enterprises can use existing SQL-trained users instead of spending time and money to train from the ground up or hunt down elusive and expensive talent. Using SQL also opens the door to existing business intelligence and visualization tools as well as existing dashboards and reports. Existing SQL applications and queries don't have to be rewritten to access Hadoop data, and the data itself does not have to be brought out of Hadoop in order to use it. In the same vein, duplicating Hadoop data is no longer necessary, assuring the cost effectiveness of the technology.

SQL ON HADOOP OPTIONS

There are many solutions attempting to deliver SQL on Hadoop capabilities. These can be divided into three camps:

- **Marketing Jobs (SQL Outside Hadoop):** Employs both Hadoop and a DBMS cluster and uses a connector to pass data between the two systems.
- **Wrapped Legacy (Mature But Non-Integrated):** These solutions take existing SQL engines and modify them so that when generating a query execution plan they can determine which parts of the query should be executed through MapReduce and SQL operators.
- **From Scratch (Integrated But Immature):** This approach builds SQL engines from the ground up to enable native SQL processing of data in HDFS while avoiding MapReduce.

SQL IN HADOOP

SQL in Hadoop is a relatively new category allowing for high-performance data enrichment and visual design capabilities without the need for MapReduce skills. Users can build and test models with data mining, data science and machine learning algorithms. Putting these into production is made simple with common SQL tools for business intelligence.

TURNING HADOOP INTO AN ANALYTICS PLATFORM

Regardless of the approach taken, with increased adoption of modern data platforms, the need to analyze large volumes of data in Hadoop and deliver scalable SQL access to that data will only grow. Modernization is critical in creating transformational value out of big data, and failing to keep up can put any enterprise's long-term position in jeopardy.

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Making Sense of Hadoop Data Lakes in a Big Data Architecture

There's no doubt that Big Data has changed the way we view and manage our IT enterprise. More and more, business leaders are recognizing that better data management enables better decision-making and more efficient operations.

Hadoop burst onto the scene a decade ago and has been gaining strength ever since. It has become synonymous with Big Data and given rise to innovation, as well as a good deal of hype. Still, many questions remain: What is Hadoop and what are its strengths and weaknesses? How will it evolve and what technologies naturally make it more effective? How can you make Hadoop enterprise ready?

One increasingly popular use of Hadoop is the Hadoop data lake. The data lake can be a promising option for enterprise-wide analytics and business intelligence. The potential benefits are clear for the lines of business, data scientists, and IT professionals alike. Data from disparate sources throughout the organization are proactively placed in the data lake. Whenever a team or data scientist wants to run analysis, the information is ready and waiting.

> Gartner predicts that by 2018, 20% of the largest 2,000 enterprises will use Hadoop as an archive repository.

As Gartner recently noted, data lakes eliminate the need to deal with dozens of independently-managed collections of data. Instead, information is combined into a single data lake. From an IT perspective, Hadoop is an ideal platform to support data lakes, given its scalability and low cost.

At first glance, data lakes seem like they could be nirvana for data scientists. From an implementation standpoint, a common approach is to incorporate an operational data store (ODS) within a data lake environment as a surefire way to deliver on the promise of increased agility through improved data accessibility.

In addition to the analytics benefits, capturing production data in the ODS ensures that organizations maintain access to production data without tasking production systems. The key to implementing a successful data lake is to first gain access to the business' data usage analytics. Enterprises need to know which data is being used, how often, and by which departments so they can make better decisions about which data to move where. Once that information is captured using new data management software, businesses need to simplify creation and maintenance of the Hadoop data lake by using an automated, high-performance data integration solution.

Following are three tips for implementing an ODS as part of a larger Hadoop data lake initiative:

1. Find a data integration tool that will keep production data up-to-date. Including production data in a data lake supply chain is only useful if that information is kept as current as the systems that generate the data. The best way to keep an ODS containing production data up-to-date is to use a solution that captures changes in the source systems as they occur and sends them to the ODS. This ensures that data scientists not only have problem-free access to the information they need, but that they can also access data that reflects the same "version of the truth" that the lines of business are working from.

2. Look for solutions that offer heterogeneous data support. Production data invariably comes from many different source systems. An automation tool that has heterogeneous data support ensures that a wide range of production systems can be used as sources for the ODS.

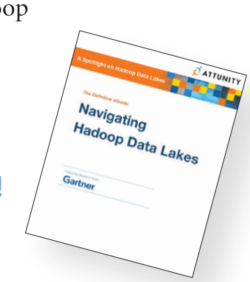
3. Seek out tools with a simple, intuitive user interface. GUI-driven designs that simplify and virtualize operations are ideal. Ease of use means that operational data stores can be created in days or hours, rather than months. That translates into rapid return on investment (ROI).

In Attunity's new Hadoop eGuide, featuring research from Gartner*, we explore the implementation and use of Hadoop data lakes. Hadoop data lakes have become increasingly more prevalent in the IT landscape because they support data scaling and accessibility. However, introducing Hadoop into an existing architecture can be a challenge.

The eGuide will also help you learn more about:

- **Gartner's view:** on Hadoop data lake trends and practices
- **Challenges:** how to overcome typical Hadoop data lake challenges
- **Tips:** which data to keep and move and best approaches for analyzing usage
- **Best practices:** using Hadoop to make data accessible enterprise-wide
- **Solutions:** for analyzing and moving data quicker and easier to/from Hadoop

Download this Hadoop Data Lakes eGuide now!



* Gartner, Predicts 2015: Managing Data Lakes of Unprecedented Enormity, John Monroe et. al., 3 Dec. 2014.

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IBM

A New Blueprint of Innovation With IBM and Apache Spark

Apache® Spark™ is an open-source cluster computing framework with in-memory processing to speed analytic applications up to 100 times faster compared to technologies on the market today. Developed in the AMPLab at UC Berkeley, Apache Spark can help reduce data interaction complexity, increase processing speed and enhance mission-critical applications with deep intelligence.

Highly versatile in many environments, Apache Spark is known for its ease of use in creating algorithms that harness insight from complex data. As a result, Apache Spark has grown quickly in popularity among developers and data scientists as an essential platform for easily integrating big data into applications, and is quickly gaining momentum with IBM clients looking to transform business decision-making.

Real-time transportation planning software from Optibus is changing the way public transport is organized. Apache Spark, together with IBM, provides a high-scalable platform for Optibus, making it easy for them to expand their software-as-a-service offerings into new markets and simplify deployment, maintenance and application development for transportation companies worldwide.

Findability Sciences, a global consulting and contextual data technology solutions company, is using IBM BigInsights and Apache Spark to help clients tap into the power of big data with greater capacity where MapReduce was not efficient before. With Apache Spark, performance improved multifold. Findability Sciences is now able to process streaming data from IoT devices and offer analytics for data in motion for things like traffic, commuters and parking.

Independence Blue Cross (IBC), the largest health insurer in the Philadelphia area, is using Apache Spark to drive product innovation and develop new services. Apache Spark quickly matured into a power tool for the development of machine-learning applications, allowing IBC researchers and partners to work together more seamlessly and get new claims and benefits apps up and out to customers faster.

IBM, NASA, and the SETI Institute are collaborating to analyze terabytes of complex deep space radio signals using Apache Spark's machine-learning capabilities in a hunt for patterns that might indicate the presence of intelligent extraterrestrial life. With Spark as a Service on Bluemix, SETI scientists are able to work with IBM to develop promising new ways to analyze signal data captured over multiple years.

Apache Spark is agile, fast and easy to use. And because it is open source, it is improved continuously by a worldwide community. Over the course of the next few months, IBM scientists and engineers will work with the Apache Spark open community to rapidly accelerate access to advanced machine-learning capabilities and help drive speed-to-innovation in the development of smart business apps.

To further accelerate open source innovation for the Spark ecosystem, IBM is taking the following actions:

- IBM will build Spark into the core of the company's analytics and commerce platforms.
- IBM's Watson Health Cloud will leverage Spark as a key underpinning for its insight platform, helping to deliver faster time to value for medical providers and researchers as they access new analytics around population health data.
- IBM will open source its breakthrough IBM SystemML machine learning technology and collaborate with Databricks to advance Spark's machine learning capabilities.
- IBM will offer Spark as a Cloud service on IBM Bluemix to make it possible for app developers to quickly load data, model it, and derive the predictive artifact to use in their app.
- IBM will commit more than 3,500 researchers and developers to work on Spark-related projects at more than a dozen labs worldwide, and open a Spark Technology Center in San Francisco for the Data Science zen labs worldwide, and open a Spark Technology Center in San Francisco for the Data Science and Developer community to foster design-led innovation in intelligent applications.
- IBM will educate more than 1 million data scientists and data engineers on Spark through extensive partnerships with AMPLab, DataCamp, MetiStream, Galvanize and Big Data University MOOC.
- By contributing SystemML, IBM will help data scientists iterate faster to address the changing needs of business and to enable a growing ecosystem of app developers to apply deep intelligence into everything.

It's not just about data access anymore. It's about building algorithms that put analytics into action. It's about changing data science and driving intelligent applications fueled by data. Combining data, design and speed, IBM and Apache Spark are creating a new blueprint of innovation.

IBM Analytics for Apache Spark's unified programming model for batch and real-time analytics, multi-language support, in-memory data processing engine, and integrated libraries such as machine-learning and graph processing, make it possible for developers, business analysts, and data scientists to incorporate intelligence into every application and better support critical-path business decision-making.

IBM

Learn more and get started at www.ibm.com/spark.